QUORAI 2.0

AI-Powered Web Research with Verifiable Agent Reliability

Team Phoenix

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GitHub	https://github.com/AnushkaSinghRajput/QUORAI
Live app (Vercel)	https://quorai-ecru.vercel.app
Failproof workspace	https://app.befailproof.ai/quoraai

1. The real-world problem

Teams, researchers, and operators increasingly ask AI systems to answer high-stakes questions from the open web. Generic chatbots often invent sources, mix outdated facts, or fail silently when a tool call loops. The result is fragile decision-making: briefs that look polished but cannot be audited, defended, or reproduced.

QUORAI attacks that gap. It turns a natural-language question into a multi-source, citation-backed research brief while recording genuine agent execution traces for reliability review — so every answer can be inspected, not just trusted.

2. What we are building

- **Live retrieval** — Tavily as primary search (Serper optional fallback).
- **Live synthesis** — OpenAI as primary LLM (Anthropic optional fallback).
- **Cited briefs** — ranked sources, citation confidence, PDF/text grounding.
- **Failproof reliability** — runtime events shipped to the QuoraAI workspace.
- **Protected vs unprotected demo** — genuine policy contrast for judges.
- **Operator console** — /reliability metrics, decisions, Cloud verify.

3. End-to-end solution (step by step)

Step A — Capture the question

A user enters a Quick Search or Deep Research query (optionally with PDF/notes). The client opens a research session and streams progress over SSE from `POST /api/research`.

Step B — Retrieve evidence

QUORAI queries the live web through Tavily, ranks domains, and merges uploaded document evidence. Tool calls are allowlisted and rate-limited inside the Failproof trace layer to stop loops and runaway execution.

Step C — Synthesize with citations

OpenAI streams a structured brief grounded in retrieved sources. Citation markers [n] are verified against the source set. Unsupported URLs and uncited claim lines are flagged; protected mode can deny weak evidence.

Step D — Observe and recover

Every run emits agent/tool/model events via the official Failproof SDK, then ships NDJSON batches to `POST /v1/events` for the QuoraAI workspace. Local reliability records capture decisions, failures, recoveries, and delivery status for the /reliability console.

Step E — Compare and decide

The /demo flow runs the same query under protected and unprotected policies so judges can see how reliability controls change outcomes using genuine execution data — not screenshots of mocked traces.

4. Architecture at a glance

Layer	Primary	Fallback / notes
Search	Tavily	Serper (optional)
LLM	OpenAI	Anthropic (optional)
Reliability	Failproof Cloud (quoraai)	Local reliability store
Hosting	Vercel Production	Next.js App Router + SSE
Auth	httpOnly sessions	Requires real AUTH_SECRET

5. How we deliver quality in production

Reliability

- No silent mock-as-live: missing Tavily/OpenAI keys return clear API errors when `ALLOW_MOCK_FALLBACK=0`.
- Loop detection, tool allowlists, execution limits, evidence and citation checks.
- Failproof traces for every research session shipped to QuoraAI Cloud.

Scalability

- Stateless Next.js API routes on Vercel with streaming responses.
- Provider chain pattern: primary + optional fallbacks without rewriting the UI.
- Rate limiting on research, demo, auth, and upload endpoints.

Maintainability

- Clear module boundaries: providers/, research/, failproof/, auth/.
- Shared list/research patterns, Zod validation, TypeScript throughout.
- Documented env template, competition guide, and verification scripts.

Security & honesty

- Server-side API keys only; secrets never committed.
- Session cookies require a real AUTH_SECRET (no hardcoded fallback).
- Credential presence diagnostics expose status without printing values.

6. Competition demo path

- Open live app: https://quorai-ecru.vercel.app
- Run a Deep Research query and show streaming stages + citations.
- Open /demo for protected vs unprotected comparison.
- Open /reliability and verify a session; then inspect QuoraAI Observe → Events.
- Repository: https://github.com/AnushkaSinghRajput/QUORAI

7. Team Phoenix

Member	Role focus
Anushka Singh Rajput	Product engineering, Vercel production, Failproof QuoraAI integration, research UX
Vedant Gupta	Reliability storytelling, competition narrative, solution framing for real-world buyers

8. Expected outcomes

QUORAI 2.0 demonstrates that AI research systems can be both useful and accountable. A decision-maker gets a cited brief in minutes; an operator gets a forensic trail of what the agent searched, synthesized, blocked, and recovered. That combination — speed with auditability — is how AI research becomes production infrastructure rather than a demo chat window.

9. Links (bookmark these)

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- **Failproof QuoraAI:** https://app.befailproof.ai/quoraai

Built with integrity — no fabricated live research claims.
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